

Assignment: Dynamic Data Management Web Page

● Objective

- Input data through a form.
- Display that data in a dynamic table.
- Delete individual entries from the table.
- Search/filter data live based on any column.
- Additionally, extend the form to include Phone Number, Branch selection, Language options, and dynamically populated State/City fields using public APIs.

● Components and Functional Requirements

1. Data Entry Form

Create a form with the following input fields:

- Name (Text)
- Age (Number)
- Email (Email format)
- Phone Number (10-digit number)
- **Branch (Radio buttons:** CSE, ECE, EEE)
- **Languages Known (Checkboxes:** e.g., English, Hindi, French, etc.)
- State (Dropdown, populated dynamically via public API)
- City (Dropdown, based on selected state, populated dynamically)

Add Button:

Validate the input fields:

- No empty values.
- Valid email format.

- Age must be a positive number.
- Phone number must be 10 digits.
- At least one language must be selected.
- Branch must be selected.
- Valid state and city must be selected.

On successful validation:

- Add the data to a dynamic display table.
- Clear the form fields.

2. Data Display Table

Display the submitted form data in a tabular format.

- **Columns:** Name, Age, Email, Phone Number, Branch, Languages, State, City

Each row should include a Delete button.

Clicking the Delete button:

- Removes the row dynamically (without page refresh).
- Optionally, shows a confirmation dialog before deletion.

3. Search Functionality

Provide a search input field above the table.

Filter table rows based on the search query.

Matching should occur on any column.

Filtering should be live and dynamic (no page reload).

If no results match, show a user-friendly "No results found" message.

● Technical Requirements

- Use HTML for page structure.
- Use CSS for styling (responsive, clean interface).
- **Use JavaScript and jQuery for:**
 - Form validation

- Row addition and deletion
- Dynamic search
- Fetching state/city data from a public API (e.g., <https://api.countrystatecity.in/>)
- Page must work dynamically—no full-page reloads on any operation.

● Bonus (Optional Enhancements)

- Show a confirmation dialog before deleting a row.
- Display inline error messages for invalid form inputs.
- Animate row addition/deletion (CSS transitions).
- Disable the Add button until the form is valid.
- Display a count of total entries shown after search/filter.

● Deliverables

A complete, fully functional web page (HTML, CSS, JS).

All source files organized and commented.

Ensure:

- Code readability and proper structure.
- Responsive design.
- Consistent UI/UX.
- No page reloads on add/delete/search.
- Functional state-city dropdown using public API.

● Evaluation Criteria

Criterion	Details
Functionality	All required features implemented correctly
Code Quality	Clean, modular, and commented code
User Experience	Form usability, responsive design, clear validation, user feedback
Use of Technologies	Effective use of jQuery, JavaScript, APIs
Dynamic Behavior	No page reloads, real-time updates, proper DOM manipulation
Optional Features	Bonus points for enhancements and UX improvements